

Two things changed. Lesson access is now decided by the server instead of the lesson-list widget, and playback progress is reported four times per lesson instead of on a rolling timer.

1. What changed in the backend

1.1 New field: is_locked

Returned on every lesson, in both the list and the detail response. A lesson is unlocked when the previous lesson in the same subject is completed. The first lesson of a subject is always unlocked. Articles are part of the sequence.

1.2 New field: checkpoints

Array of second offsets at 25%, 50%, 75% and 100% of the duration. These are the only points at which the app should report progress.

```
1 "duration_seconds": 7200,
2 "checkpoints": [1800, 3600, 5400, 7200]
```

Empty for articles and for locked lessons. On very short clips two quarters can round to the same second, so the array may hold fewer than four values — read its length.

1.3 Locked lessons are refused, not just flagged

These three return 403 with a localized message when the lesson is locked:

- GET /user/content/{id}
- POST /user/content/{id}/access-token
- POST /user/content/{id}/progress

The access-token refusal is the one that matters — no token means no playable URL.

1.4 article_body removed from the list

The list returns card data only. Article text comes from GET /user/content/{id}, and only when that lesson is unlocked.

1.5 Seek guard

POST /progress returns 422 if the reported position moved further than real elapsed time allows. The body carries allowed_position_seconds.

1.6 Not changed

Media URLs are still never returned in any content response — audio and video remain reachable only through the access token. Posting position_seconds = 0 against an article still marks it complete, so the read button needs no new endpoint.

2. What the app has to do

2.1 Delete the periodic timer

The current build posts to /progress every 3–5 seconds. Remove it and every call site that is not a checkpoint.

2.2 Count locally in Hive

Keep the one-second tick and persist it to a Hive box keyed by content id: content_id, position_seconds, watched_delta_seconds, next_checkpoint_index. Write on every tick, send nothing in between. Between checkpoints the player produces zero requests.

2.3 Report at the four checkpoints

When the local position reaches checkpoints[next_checkpoint_index], send one request, increment the index, reset the delta. In order — never checkpoint 3 before checkpoint 2. Do not compute the offsets yourself.

```
1 POST /api/user/content/{id}/progress
2   position_seconds      int    the checkpoint value
3   watched_delta_seconds int    seconds actually played since the
   last report
```

Status

IN PROGRESS

Action required

MA Mohammad AlKhat...

Not scheduled

Assigned to

MA Mohammad AlKhat...

All fields

Time Reports

> Users

MA AS

`watched_delta_seconds` is played duration, not the difference between positions. Send it every time — without it the server derives attendance from position movement and credits a rewatch as zero.

Response — use these values for the UI rather than the local counter:

```
1 {
2   "allowed_position_seconds": 1800,
3   "max_position_seconds": 1800,
4   "watched_seconds": 1800,
5   "progress_percent": 25,
6   "is_completed": false
7 }
```

2.4 Send the last checkpoint when the video ends

The player stops on its own at the end. Do not gate the send on `isPlaying` — that is what previously left lessons stuck below completion.

2.5 Cap the seek bar at `max_position_seconds`

Rewinding and rewatching are fine. Dragging forward past the watched point must be blocked in the UI, not corrected after the fact.

2.6 Treat 422 as a correction

Seek to `allowed_position_seconds` and carry on. No dialog, no error state, no retry of the same value.

2.7 Draw the lock from `is_locked`

Delete the ordering and completion comparison in the lesson list widget. One source of truth.

2.8 Do not sign the user out on 403

Show the `message` from the response body — it is already localized, do not hardcode a string. Check that the global Dio interceptor does not treat 403 as an expired session.

2.9 Open lessons through the detail endpoint

`GET /user/content/{id}` returns `article_body`, `checkpoints` and `attachments`. The list no longer carries the body.

2.10 Article read button

At the bottom of the article screen, only when `type = "article"`. Disabled or replaced by a read indicator when `progress.is_completed` is already true.

```
1 POST /api/user/content/{id}/progress
2   position_seconds = 0
```

Reload the subject lesson list on success so the next lesson opens.

2.11 Refresh lock state after any completion

`is_locked` is not cacheable. Reload the list whenever a lesson finishes or an article is marked read.

2.12 Resume, do not replay

On reopening a lesson, restore position and checkpoint index from Hive and continue. A checkpoint crossed while the app was closed is not re-sent — the server rejects it as a skip. Clear the Hive entry after the fourth checkpoint is acknowledged.

3. Fields

Field	Type	List	Detail	Notes
<code>is_locked</code>	bool	yes	yes	New
<code>checkpoints</code>	int[]	yes	yes	New. Empt articles and lessons
<code>article_body</code>	string	—	yes	Removed from list
<code>attachments</code>	array	—	yes	Metadata changed
<code>subject_name</code>	string	—	yes	Unchanged
<code>progress</code>	object null	yes	yes	Unchanged

4. Worked example — a two-hour video

Report	<code>position_seconds</code>	<code>watched_delta_seconds</code>	<code>progress_percent</code>	<code>is_completed</code>
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1st	1800	1800	25	false
2nd	3600	1800	50	false
3rd	5400	1800	75	false
4th	7200	1800	100	true

The delta equals a full quarter only when the student watched it straight through. A pause makes it smaller, a rewatch makes it larger. Report what the counter actually accumulated. Never send two reports back to back to catch up — the server measures each against real elapsed time and rejects the second.

5. Edge cases

Situation	Expected behaviour
Clip shorter than four seconds	Fewer than four checkpoints. Iterate the array, do not index a fixed <code>[3]</code> .
Lesson has no duration	<code>checkpoints</code> is empty. No reporting.
App closed mid-quarter	Unreported seconds are lost. Acceptable — do not buffer and flush later.
Student rewatches an earlier part	Report the real delta. <code>max_position_seconds</code> does not move backwards.
Access token expires mid-lesson	Request a new one. Hive state is untouched.
Locked lesson opened by deep link	403 from the detail endpoint. Show the message, stay on the list.

6. Test data

Seeded under `category_subject_id = 20` :

Content	Type	State	Unlocked by
10	video, 10s, checkpoints <code>[2,5,7,10]</code>	unlocked	First in subject
11	article	locked	Completing 10
12	audio, 12s, checkpoints <code>[3,6,9,12]</code>	locked	Marking 11 read

Lesson 10 runs ten seconds, so the whole four-report cycle is observable in under a minute.

Show less ↑

Messages

AS

AlMouayad Shwin created the task

6 days

AS

AlMouayad Shwin

6 days

Action required by Mohammad AlKhateeb

Mohammad AlKhateeb

5 days

Changed status to **IN PROGRESS**

MA

Action is required by **Mohammad AlKhateeb** (6 days)

Reply / Comment